Physics-Infused Reduced-Order Modeling for Analysis of Multi-Layered Hypersonic Thermal Protection Systems With Unknown Thermal Contact Resistances

Abstract

This work presents a systematic design and implementation of the physics-infused reduced-order modeling (PIROM) framework for optimal inference and reduced-order modeling of hypersonic TPS with unknown thermal contact resistances (TCRs). The PIROM framework is designed to leverage the strengths of both physics-based and data-driven modeling paradigms, by infusing physics-based models with data-driven components to enhance their predictive capabilities and generalizability. Inference heat conduction problems (IHCPs) are solved using the PIROM framework to infer TCRs from internal temperature measurements, while providing accurate predictions of transient TPS temperature dynamics under varying boundary conditions and material properties. Particularly, experimental limitations are considered in the design of the PIROM framework, including noisy observables and thermocouple placement limitations. The performance of the PIROM framework is benchmarked against traditional physics-based modeling approaches (e.g, one-dimensional FEM) and data-driven surrogate models (e.g., neural ordinary differential equations) for both inferring contact conductances and transient TPS temperature dynamics.

1 Introduction

What is the importance of hypersonic TPS? Hypersonic thermal protection systems (TPS) play a critical role in safeguarding hypersonic vehicles as they are exposed to extreme aerodynamic heating during flight. Therefore, accurate modeling and prediction of TPS behavior become essential for designing effective thermal protection strategies. Nonetheless, the highly non-linear fluid-thermal-structural interactions, temperature-dependent material property variations, and uncertainties in thermal contact resistances (TCR) at the interfaces of the TPS layers, pose significant challenges for traditional modeling approaches such as finite-element methods (FEM), as well as reduced-order (ROM) and surrogate (SM) modeling techniques.

What is the new problem that we want to solve? Particularly, a critical artifact in transient reduced-order modeling of hypersonic TPS is the presence of TCRs. Due to manufacturing limitations, when two solids are brought together to form an interface, the true solid-to-solid contact area between them is generally a small fraction of the apparent area over which they meet [Minges1970]. The net effect of imperfect contact is the formation of a temperature discontinuity at the interface that is in general challenging to predict, and that significantly affects the heat transfer across the interface. The identification of TPS properties from internal temperature measurements belongs to a class of *inverse heat conduction problem* (IHCP) [Deng2026], which are known to be ill-posed and sensitive to measurement noise, and thus require careful regularization and robust inference techniques for accurate estimation of TCRs. The challenges posed by IHCPs can introduce significant uncertainties in transient temperature predictions, and thus compromise the reliability of ROMs for design, analysis, and optimization of hypersonic TPS [Fang2023]. Therefore, there is a critical need for *efficient*, *accurate*, and *generalizable* modeling approaches of TPS that can infer TCRs, while providing reliable predictions of internal TPS transient temperaures under varying boundary conditions and material properties.

What has been done in the literature for inference and simulation? Attempts to modeling hypersonic TPS with varying TCRs have taken various forms in the literature, ranging from traditional deterministic physics-based modeling approaches, to probabilistic approaches, and more recently, to scientific machine learning (SciML) techniques that leverage the strengths of both physics-based and data-driven modeling paradigms. For instance, in Ref. [Fang2023], a Lattice Boltzmann Method (LBM) is used to model the transient heat transfer and solid-liquid phase change in a hypersonic TPS, where the TCRs are assumed known, and are shown to significantly delay the melting points of phase changing materials. In Ref. [Zhang2024], authors take one step further by treating the TCRs as unknown parameters, and a finite-element thermo-mechanical model of a three-dimensional reusable heat-pipe cooled thermal protection system is developed to perform Levenberg-Marquardt gradient-based inference of internal TCRs using boundary temperature measurements. A similar approach is adopted in Ref. [Brociek2024], where a finite-difference one-dimensional model of a three-layered TPS is developed to solve the forward problem, and a Levenberg-Marquardt for a Tikhonov regularized optimization is used to solve the inverse problem of inferring TCRs from internal temperature measurements.

Probabilistic approaches to solving the inverse problem of inferring TCRs have also been proposed in the literature. For instance, in Ref. [Gao2026], a Bayesian Network (BN) with sequential importance resample (SIR) is leveraged for proabilistic inversion of TCRs, where the BN is implemented by leveraging a finite-element model of the multi-layered TPS for forward simulations. A BN is also leveraged in Ref. [Deng2026] to infer TCRs, where a neural network is used as a surrofate model to predict internal temperature dynamics due to the computational cost of the FEM-based forward model.

The use of scientific machine learning (SciML) techniques for modeling hypersonic TPS with unknown TCRs has limited exploration in the literature.

.... Other X approaches such as those in Refs.[X] ...

While these methods offer a high-fidelity representation of the underlying physics, highly refined FEMs result in large-scale systems that are computationally expensive to solve, and thus are not suitable for design, analysis, and optimization of hypersonic TPS.

Other data-driven approaches such as those in Refs.[X] ..., leverage Bayesian Networks (BNs) to learn mappings from boundary conditions and thermocop

machine learning techniques to learn the underlying dynamics of the TPS system directly from data, while inferring the TCRs. For instance, in Refs. [X]...

While these methods offer computational efficiency, they often require large amounts of data for training, and may not generalize well to unseen conditions, such as varying boundary conditions and material properties. Moreover, they may not provide insights into the underlying physics of the system, which can be critical for design and optimization.

What is the present work doing? This work presents a systematic design, implementation, and benchmarking of the PIRM for modeling hypersonic TPS with unknown TCRs under experimental constraints such as noisy observables and thermocouple placement limitations.

Objectives of the work The objectives of this work are to,

1. Demonstrate the systematic application of PIROM for modeling hypersonic TPS with unknown contact resistances, emphasizing on the simultaneous inference of material properties and hidden dynamics with limited data.

2. Benchmark the performance of PIROM against traditional physics-based modeling approaches (e.g, one-dimensional FEM) and data-driven surrogate models (e.g., neural ordinary differential equations) for both inferring contact conductances and transient TPS temperature dynamics.

3. Provide insights into the optimal design of physics-based models for inference of contact conductances, including the resolution of the physics-based model, noise levels, and constraints with the placement of thermocouples.

2 Modeling of Hypersonic Protection Systems

The following discussion introduces the governing equations in the heat conduction problem for hypersonic TPS with unknown contact resistances. Moreover, three different physics-based approaches of decreasing fidelity are presented for solving the governing equations: (i) a fine three-dimensional FEM termed the *full-order model* (FOM), (ii) a one-dimensional FEM with linear elements termed the *one-dimensional FEM* (1D-FEM), and (iii) a lumped-mass representation of the TPS termed the *lumped capacitance model* (LCM). The governing equations and their three solutions strategies are presented next.

2.1 Governing Equations

Consider the generic domain $\Omega \subset \mathbb{R}^d$ with $d = 1, 2, 3$ as show in Fig. X. A heat flux q_b (i.e., Neumann boundary condition) is prescribed on boundary Γ_q, while a temperature T_b (i.e., Dirichlet boundary condition) is prescribed on boundary Γ_T, and $\partial\Omega = \Gamma_q \cup \Gamma_T$. The transient heat conduction dynamics is governed by the following energy PDE,

$$
\begin{aligned}
\rho c_p \partial_t T - \nabla \cdot (\mathbf{k}(T)\nabla T) &= 0,\ x \in \Omega && \text{(1a)}\\
-\mathbf{k}(T)\nabla T \cdot \mathbf{n} &= q_b(x,t),\ x \in \Gamma_q && \text{(1b)}\\
T(x,t) &= T_b(x,t),\ x \in \Gamma_T && \text{(1c)}\\
T(x,0) &= T_0(x),\ x \in \Omega && \text{(1d)}
\end{aligned}
$$

where the density ρ is constant, while the heat capacity c_p and thermal conductivity $\mathbf{k} \in \mathbb{R}^{d\times d}$ are temperature-dependent. The TPS domain is divided into N non-overlapping components $\{\Omega_i\}_{i=1}^N$, as illustrated in Fig. [X] for $N = 2$, where the thermal contact resistance (TCR) at the interfaces between adjacent components is treated as an unknown parameter. The presence of TCRs introduces a temperature discontinuity at the interfaces, which significantly affects the heat transfer across the interfaces, and thus the overall thermal response of the system.

2.2 Full-Order Model

The FOM is obtained by spatially discretizing the governing PDE using a Discontinuous Galerkin (DG) FEM to result in a high-dimensional system of ordinary differential equations (ODEs) that captures the transient temperature dynamics of the TPS with high fidelity. As in previous work

[VargasVenegas2026], the DG-FEM is chosen to describe the FOM as it: (i) provices a systematic framework for modeling multi-layered TPS with complex geometries and material properties, (ii) provides a natural framework for incorporating TCRs at the interfaces, and (iii) is amenable to systematic model reduction via coarse-graining. In Sec. [X] the FOM numerical solutions aer performed using standard FEM instead, and the equivalence between DG and standard FEM is noted upon their convergence.

Consider a DG-FEM spatial discretization of the governing PDE, where the domain Ω is partitioned into M non-overlapping elements $\{E_i\}_{i=1}^{M}$, and the temperature field is approximated by a set of local basis functions $\boldsymbol{\phi}_i$ defined on each element E_i.

The DG-FEM states are given by,

$$\mathbf{u}(t) = [\mathbf{u}_1(t), \mathbf{u}_2(t), \ldots, \mathbf{u}_M(t)]^\top \in \mathbb{R}^{MP} \tag{2a}$$

$$\mathbf{w}(t) = [\mathbf{w}_1(t), \mathbf{w}_2(t), \ldots, \mathbf{w}_M(t)]^\top \in \mathbb{R}^{KP} \tag{2b}$$

where $\mathbf{w}$ corresponds to a subset of the DG-FEM states that are adjacent to the interfaces between components, and thus are used to compute the temperature jumps across the interfaces. Clearly, $M \gg K$, since the number of elements is much larger than the number of elements adjacent to the interfaces, nonetheless, KP remains much larger than the $N_C + N_\Gamma$ which is the total number of components and interfaces.

We can introduce the following condition to express the average interface temperature as a function of the element-wise temperature states (the average),

$$\int_{\Gamma_j \cap \partial E_m} \boldsymbol{\phi}_p^j d\Gamma = 0, \quad p > 2 \tag{3}$$

2.3 One-Dimensional Model

This approach follows the classical FEM approach with a linear basis function to capture the through-thickess spatially-varying temperature distributions inside each component of the TPS. This modeling simplification is commonly used in the literature for modeling heat conduction in multi-layered TPS, since under hypersonic thermal loading, the temperature gradient along the thickness direction of the structure is significantly higher than the in-plane temperature gradients, and thus the through-thickness temperature distribution dominates the overall thermal response of the system [Gao2026].

As with the FOM, the TCR introduces additional considerations in the formulation of the 1D-FEM.

3 Lumped Capacitance Model with Imperfect Contacts

This work leverages the *lumped capacitance model* (LCM), a first-principle physics-based model of the TPS that captures the transient component-wise average temperature of the TPS [Incropera2011]. Nonetheless, the presence of TCRs at the interfaces introduces additional modeling considerations since the temperature drop across an interface includes, in addition to the bulk drop across the interface, a contribution from the contact resistance. The following discussion presents the formulation of the LCM for modeling the transient thermal response of the TPS with unknown TCRs, and the separation of dynamics into *bulk* and *contact* contributions.

Bulk Dynamics Leveraging our previous work in Ref. [VargasVenegas2026], the LCM is leveraged in this work as the physics-based component in the PIROM. The LCM may be derived at the component level from a point of view of energy conservation, and is written as,

$$\int_{\Omega_i} \rho c_{p,i}(\bar{u})\dot{\bar{u}}_i d\Omega = \sum_{j\in\mathcal{N}_i} \int_{\Gamma_{ij}} q_{ij} d\Gamma + \int_{\Gamma_{iq}} q_b d\Gamma + \int_{\Gamma_{iT}} q_T d\Gamma \tag{4}$$

where q_{ij} and Γ_{ij}, are the heat flux and contact interface between neighboring components i and j, q_b and Γ_{iq} are the heat flux and boundary where Neumann conditions are applied, and q_T and Γ_{iT} are the heat flux and boundary where Dirichlet conditions are applied. The set $\mathcal{N}_i$ corresponds to the neighboring components of component i, which are defined as those components that share an interface with component i. In the case of a perfect contact between components, the interfacing heat flux q_{ij} is written as $q_{ij} = G_{ij}(\bar{\mathbf{u}})(\bar{u}_i - \bar{u}_j)$ where $G_{ij}(\bar{\mathbf{u}})$ is the effective conductance of the interface between components i and j, which may be a function of the average temperature $\bar{\mathbf{u}}$. However, in the presence of TCRs, the heat flux across the interface includes a contribution from the contact resistance in the form of a temperature drop across the interface, as explained next.

Contact Dynamics Consider Fig. [X], where an interface between two layers of the TPS shown in Fig.[X] is magnified to depict the governing heat transfer mechanisms. The variables θ_i^- and θ_i^+ represent the averaged interfacial temperatures, immediately to the left and to the right of the interface, respectively, while z_i^- and z_i^+ represent the thermocouple measurements at distances δ_i^- and δ_i^+ to the left and right of the interface, respectively. Additionally, R_i^-, R_i^+, and $R_{x,i}$ correspond to the bulk thermal resistance of the left and right layers, and the contact resistance at the interface, respectively. The total temperature drop across the interface is therefore the sum of the bulk temperature drop and the contact temperature drop, i.e.,

$$\underbrace{\bar{u}_i - \bar{u}_{i+1}}_{\text{total lumped drop}} = \underbrace{\left(\bar{u}_i - \theta_i^-\right) + \left(\theta_i^+ - \bar{u}_{i+1}\right)}_{\text{bulk drop}} + \underbrace{\left(\theta_i^- - \theta_i^+\right)}_{\text{contact drop}} \tag{5}$$

hence, the bulk drop becomes,

$$\left(\bar{u}_i - \theta_i^-\right) + \left(\theta_i^+ - \bar{u}_{i+1}\right) = \left(\bar{u}_i - \bar{u}_{i+1}\right) - \Delta\theta_i \tag{6}$$

Therefore, at an interface $i \in \mathcal{I}$ with a contact resistance, the heat flux between neighboring components may be written as,

$$q_{ij} = G_{ij}(\bar{\mathbf{u}})\Big(\left(\bar{u}_i - \bar{u}_j\right) - \Delta\theta_i\Big) \tag{7}$$

Consider the lumped-mass average temperature state,

$$\mathbf{u} = [u_1, u_2, u_3, u_4]^T \in \mathbb{R}^N \tag{8}$$

and temperature-dependent lumped-capacitance matrix,

$$A(\mathbf{u}) = \begin{bmatrix} a_1(u_1) & 0 & 0 & 0 \\ 0 & a_2(u_2) & 0 & 0 \\ 0 & 0 & a_3(u_3) & 0 \\ 0 & 0 & 0 & a_4(u_4) \end{bmatrix} \tag{9}$$

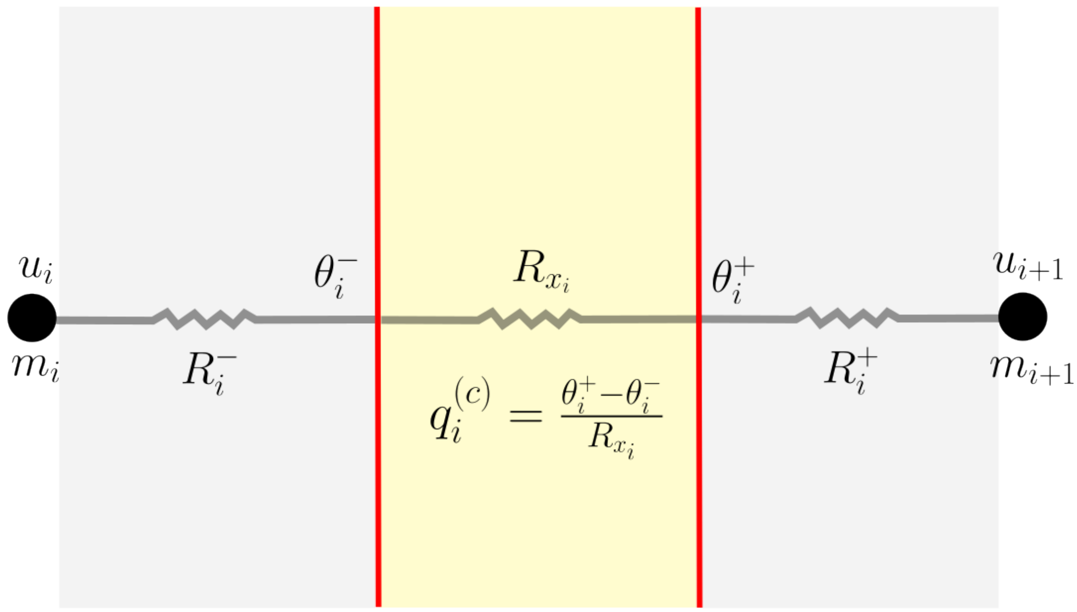


Figure 1: Schematic of the interface between two layers of the TPS, where the temperature drop across the interface is decomposed into a bulk drop and a contact drop.

Let the external Neumann forcing be applied to the first lumped mass,

$$\mathbf{f}(t) = \begin{bmatrix} f(t) \\ 0 \\ 0 \\ 0 \end{bmatrix} \tag{10}$$

For interfaces $i = 1, 2, 3$, define the known *bulk resistances*

$$S_i(\mathbf{u}) = R_i^-(\mathbf{u}) + R_i^+(\mathbf{u}) \tag{11}$$

and the unknown contact resistance,

$$R_{xi} > 0, \qquad g_{xi} := \frac{1}{R_{xi}}. \tag{12}$$

The classical lumped-capacitance model with effective series resistances is,

$$A\left(\mathbf{u}\right)\dot{\mathbf{u}} = B\left(\mathbf{u}\right)\mathbf{u} + \mathbf{f}(t) \tag{13}$$

where,

$$B\left(\mathbf{u}\right) = \sum_{i=1}^{N-1} \frac{1}{S_i(\mathbf{u}) + R_{xi}} \mathbf{K}_i \tag{14}$$

with standard nearest-neighbor diffusion stencils,

$$\mathbf{K}_1 = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \mathbf{K}_2 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \mathbf{K}_3 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \tag{15}$$

The effective heat flux through interface i is therefore,

$$q_i = \frac{u_{i+1} - u_i}{S_i(\mathbf{u}) + R_{xi}} \tag{16}$$

where the temperature-drop convention is always taken as,

$$\Delta u_i = u_{i+1} - u_i \tag{17}$$

3.1 Total drop = bulk drop + contact drop

To separate bulk and contact contributions, introduce interface temperatures

$$\theta_i^+, \qquad \theta_i^-, \qquad i = 1, 2, 3, \tag{18}$$

where θ_i^- and θ_i^+ are the temperatures immediately to the left and right of contact i, respectively. Define the contact temperature jumps as,

$$\Delta\theta_i := \theta_i^+ - \theta_i^-, \qquad i = 1, 2, 3 \tag{19}$$

The total lumped-to-lumped temperature drop across interface i,

$$u_{i+1} - u_i \tag{20}$$

This total lumped-to-lumped temperature drop is the sum of the bulk drop across the interface and the contact jump across the interface,

$$\underbrace{u_{i+1} - u_i}_{\text{total drop}} = \underbrace{\left(\theta_i^- - u_i\right) + \left(u_{i+1} - \theta_i^+\right)}_{\text{bulk drop}} + \underbrace{\left(\theta_i^+ - \theta_i^-\right)}_{\text{contact drop}}. \tag{21}$$

Hence,

$$\underbrace{\left(\theta_i^- - u_i\right) + \left(u_{i+1} - \theta_i^+\right)}_{\text{bulk drop}} = \underbrace{\left(u_{i+1} - u_i\right)}_{\text{total drop}} - \underbrace{\Delta\theta_i}_{\text{contact drop}} \tag{22}$$

$$\text{bulk drop across interface } i = (u_{i+1} - u_i) - \Delta\theta_i. \tag{23}$$

Thus the lifted state is

$$x = \begin{bmatrix} u \\ \Delta\boldsymbol{\theta} \end{bmatrix} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ \Delta\theta_1 \\ \Delta\theta_2 \\ \Delta\theta_3 \end{bmatrix} \in \mathbb{R}^7, \qquad \Delta\boldsymbol{\theta} = \begin{bmatrix} \Delta\theta_1 \\ \Delta\theta_2 \\ \Delta\theta_3 \end{bmatrix}.$$

3.2 Heat-flux Continuity and DAE Formulation

Using the adopted convention, define the bulk and contact heat fluxes as

$$q_i^{(b)} = \frac{(u_{i+1} - u_i) - \Delta\theta_i}{S_i(u)}, \qquad q_i^{(c)} = \frac{\Delta\theta_i}{R_{xi}} = g_{xi}\Delta\theta_i. \tag{24}$$

Heat-flux continuity across each interface requires

$$q_i^{(b)} = q_i^{(c)}, \qquad i = 1, 2, 3, \tag{25}$$

which yields the algebraic constraints

$$\frac{(u_{i+1} - u_i) - \Delta\theta_i}{S_i(u)} - g_{xi}\Delta\theta_i = 0, \qquad i = 1, 2, 3. \tag{26}$$

Lumped-temperature dynamics

The four lumped energy balances yield,

$$a_1(u_1)\dot{u}_1 = f(t) - q_{12}^{(b)} \tag{27a}$$

$$a_2(u_2)\dot{u}_2 = q_{12}^{(b)} - q_{23}^{(b)} \tag{27b}$$

$$a_3(u_3)\dot{u}_3 = q_{23}^{(b)} - q_{34}^{(b)} \tag{27c}$$

$$a_4(u_4)\dot{u}_4 = q_{34}^{(b)} \tag{27d}$$

Substituting using the bulk heat fluxes yields,

$$a_1(u_1)\dot{u}_1 = f(t) - \frac{(u_2 - u_1) - \Delta\theta_1}{S_1(u)} \tag{28a}$$

$$a_2(u_2)\dot{u}_2 = \frac{(u_2 - u_1) - \Delta\theta_1}{S_1(u)} - \frac{(u_3 - u_2) - \Delta\theta_2}{S_2(u)} \tag{28b}$$

$$a_3(u_3)\dot{u}_3 = \frac{(u_3 - u_2) - \Delta\theta_2}{S_2(u)} - \frac{(u_4 - u_3) - \Delta\theta_3}{S_3(u)} \tag{28c}$$

$$a_4(u_4)\dot{u}_4 = \frac{(u_4 - u_3) - \Delta\theta_3}{S_3(u)} \tag{28d}$$

Matrix form of DAE

Define the difference matrix,

$$\mathbf{D} = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}, \ \mathbf{D}u = \begin{bmatrix} u_2 - u_1 \\ u_3 - u_2 \\ u_4 - u_3 \end{bmatrix} \tag{29}$$

Define the bulk temperature stencils,

$$\mathbf{K}_1 = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \ \mathbf{K}_2 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \ \mathbf{K}_3 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \tag{30}$$

and the bulk jump-coupling stencils,

$$\mathbf{L}_1 = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \mathbf{L}_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \mathbf{L}_3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & -1 \end{bmatrix} \tag{31}$$

Then,

$$A(u)\dot{u} = \sum_{i=1}^{N-1} \frac{1}{S_i(u)} \Big(\mathbf{K}_i u + \mathbf{L}_i \Delta\boldsymbol{\theta} \Big) + \mathbf{f}(t) \tag{32}$$

and the algebraic constraints for heat flux continuity are written as,

$$\mathrm{diag}\left(\frac{1}{S_1(u)}, \frac{1}{S_2(u)}, \frac{1}{S_3(u)} \right) \mathbf{D}u - \left[\mathrm{diag}\left(\frac{1}{S_1(u)}, \frac{1}{S_2(u)}, \frac{1}{S_3(u)} \right) + G_x \right] \Delta\boldsymbol{\theta} = 0 \tag{33}$$

where

$$G_x = \begin{bmatrix} g_{x1} & 0 & 0 \\ 0 & g_{x2} & 0 \\ 0 & 0 & g_{x3} \end{bmatrix} \tag{34}$$

Equation (33), along with the lumped-temperature dynamics, defines a DAE system for the lifted state $x = [u, \Delta\boldsymbol{\theta}]$.

3.3 DAE index reduction

The DAE is index-reduced by differentiating the heat-flux continuity constraints with respect to time, which, for each interface i, yields,

$$\frac{d}{dt} \left[\frac{(u_{i+1} - u_i) - \Delta\theta_i}{S_i(u)} - g_{xi} \Delta\theta_i \right] = 0 \tag{35}$$

The contact conductances g_{xi} are assumed constant, therefore,

$$\big(1 + g_{xi} S_i(u)\big) \Delta\dot{\theta}_i = \big(\dot{u}_{i+1} - \dot{u}_i\big) - \frac{(u_{i+1} - u_i) - \Delta\theta_i}{S_i(u)} \dot{S}_i(u) \tag{36}$$

Using the original algebraic constraint

$$\frac{(u_{i+1} - u_i) - \Delta\theta_i}{S_i(u)} = g_{xi} \Delta\theta_i, \tag{37}$$

the index-reduced dynamics become,

$$\boxed{\Delta\dot{\theta}_i = \frac{\dot{u}_{i+1} - \dot{u}_i - g_{xi} \Delta\theta_i \, \dot{S}_i(u)}{1 + g_{xi} S_i(u)}, \qquad i = 1, 2, 3} \tag{38}$$

Define

$$M(u; R_x) = \begin{bmatrix} 1 + g_{x1} S_1(u) & 0 & 0 \\ 0 & 1 + g_{x2} S_2(u) & 0 \\ 0 & 0 & 1 + g_{x3} S_3(u) \end{bmatrix} \tag{39}$$

and,

$$\dot{S}(u) = \begin{bmatrix} \dot{S}_1(u) \\ \dot{S}_2(u) \\ \dot{S}_3(u) \end{bmatrix}, \qquad \Delta\boldsymbol{\theta} \odot \dot{S}(u) = \begin{bmatrix} \Delta\theta_1 \dot{S}_1(u) \\ \Delta\theta_2 \dot{S}_2(u) \\ \Delta\theta_3 \dot{S}_3(u) \end{bmatrix} \tag{40}$$

Then, the index-reduced dynamics for the contact jumps can be written in matrix form as,

$$M(u; R_x)\Delta\dot{\theta} = D\,\dot{u} - G_x\big(\Delta\boldsymbol{\theta} \odot \dot{S}(u)\big) \tag{41}$$

Thus, the index-reduced lifted system is given by the coupled ODEs,

$$A(u)\dot{u} = \sum_{i=1}^{N-1} \frac{1}{S_i(u)}\Big(\mathbf{K}_i u + \mathbf{L}_i \Delta\boldsymbol{\theta}\Big) + \mathbf{f}(t) \tag{42a}$$

$$M(u; R_x)\Delta\dot{\theta} = \mathbf{D}\dot{u} - G_x\big(\Delta\boldsymbol{\theta} \odot \dot{S}(u)\big) \tag{42b}$$

4 Physics-Infused Reduced-Order Modeling

The formulation of PIROM for TPS starts by mathematically connecting the FOM, i.e., the DG-FEM, to the RPM, i.e., the LCM. This results presented in this section leverage the previous work in Ref. X, where a coarse-graining procedure is developed to systematically derive the LCM from the DG-FEM. Such coarse-graining procedure pinpoints the missing dynamics in the LCM when compared to DG-FEM, enabling the systematic design of a memory closure to account for residual dynamics associated with discarded spatial resolution.

4.1 Deriving the Reduced-Physics Model via Coarse-Graining

As demonstrated in previous work [VargasVenegas2026], the LCM can be viewed as a special case of the DG-FEM, where the mesh partition is extremely coarse, and the trail and test functions are piece-wise constants. In this setting, the LCM becomes a coarse zeroth-order DG-FEM with the inverse of thermal resistance chosen as the element-wise penalty factors. In this work, the presence of unknown TCRs motivates the design of a coarse-graining procedure that explicitly accounts for the presence of interfaces with unknown contact conductances, which are not captured by the LCM.

4.1.1 Coarse-Graining of States

Consider a DG-FEM as in Eq. [X] for M elements, and a LCM as in Eq. [X] for N components with N_Γ interfaces; clearly, $M \gg N$. Let $\mathcal{V}_j$, $\mathcal{F}_i^-$, and $\mathcal{F}_i^+$ be the sets of indices for the elements contained in component j, and the elements adjacent to interface i on the negative and positive sides. Formally,

$$\mathcal{V}_j = \{i : E_i \subset \Omega_j\} \tag{43a}$$

$$\mathcal{F}_i^- = \{m : \Gamma_i \cap \partial E_m \neq \emptyset, E_m \subset \Omega_{j^-}\} \tag{43b}$$

$$\mathcal{F}_i^+ = \{n : \Gamma_i \cap \partial E_n \neq \emptyset, E_n \subset \Omega_{j^+}\} \tag{43c}$$

Therefore, the volume and left-and-right interface averages for component j and interface k are given by,

$$\bar{u}_j = \frac{1}{|\Omega_j|}\sum_{i\in\mathcal{V}_j}\int_{E_i}\boldsymbol{\phi}^\top \mathbf{u}_i d\Omega_i = \frac{1}{|\Omega_j|}\sum_{i\in\mathcal{V}_j}\boldsymbol{\varphi}_i^{j\top}\mathbf{u}_i,\ j=1,\dots,N \tag{44a}$$

$$\bar{\theta}_k^-(t) = \frac{1}{|\Gamma_k|}\sum_{m\in\mathcal{F}_k^-}\int_{\Gamma_k\cap\partial E_m}\boldsymbol{\phi}_m^\top \mathbf{w}_m d\Gamma_k,\ k=1,\dots,N_\Gamma \tag{44b}$$

$$\bar{\theta}_k^+(t) = \frac{1}{|\Gamma_k|}\sum_{n\in\mathcal{F}_k^+}\int_{\Gamma_k\cap\partial E_n}\boldsymbol{\phi}_n^\top \mathbf{w}_n d\Gamma_k,\ k=1,\dots,N_\Gamma \tag{44c}$$

where $|\Omega_j|$ and $|\Gamma_k|$ are the area ($d=2$) or volume ($d=3$) of component j, and the area of the interface k between two different components, respectively. The orthogonal basis functions are defined as $\boldsymbol{\varphi}_i^j = [1,0,\dots,0]\in\mathbb{R}^P$.

Therefore, it follows that,

$$\bar{\theta}_j^-(t) = \sum_{i\in\mathcal{F}_j^-}\frac{|\Gamma_i|}{|\Gamma_j|}\bar{u}_i = \sum_{i\in\mathcal{F}_j^-}\frac{|\Gamma_i|}{|\Gamma_j|}\boldsymbol{\varphi}_i^{j\top}\mathbf{u}_i(t) \tag{45a}$$

Note that,

$$\delta\theta_i^-(\mathbf{u},t) = T_i^-(\mathbf{u},t) - \bar{\theta}_i^-(\mathbf{u},t) \tag{46a}$$

$$\delta\theta_i^+(\mathbf{u},t) = T_i^+(\mathbf{u},t) - \bar{\theta}_i^+(\mathbf{u},t) \tag{46b}$$

so that,

$$\Delta\theta_i(\mathbf{u},t) = \left(\bar{\theta}_i^-(t) - \bar{\theta}_i^+(t)\right) + \left(\delta\theta_i^-(\mathbf{u},t) - \delta\theta_i^+(\mathbf{u},t)\right) \tag{47a}$$

$$\bar{\theta}_i^-(t) = C_i^-\mathbf{u}(t) \tag{47b}$$

$$\bar{\theta}_i^+(t) = C_i^+\mathbf{u}(t) \tag{47c}$$

$$\Delta\bar{\boldsymbol{\theta}}(t) = \boldsymbol{\Psi}^+\mathbf{u}(t) \tag{47d}$$

where,

$$\boldsymbol{\Psi}^+ = \begin{bmatrix} C_1^- - C_1^+ \\ C_2^- - C_2^+ \\ \vdots \\ C_{N-1}^- - C_{N-1}^+ \end{bmatrix} \tag{48}$$

Therefore, the coarse-grained states for element and interface i are given by

$$\mathbf{u}_i(t) = \sum_{k=1}^{N}\boldsymbol{\varphi}_k^{i\top}\bar{u}_k(t) + \delta\mathbf{u}_i(t),\ i=1,\dots,N \tag{49a}$$

$$\Delta\theta_i(\mathbf{u},t) = \sum_{k=1}^{N-1}\psi_k^i\Delta\bar{\theta}_k(t) + \delta\Delta\theta_i(\mathbf{u},t),\ i=1,\dots,N_\Gamma \tag{49b}$$

4.1.2 Coarse-Graining of Dynamics

Next, the coarse-graining operator for the LCM dynamics is selected as in previous work [VargasVenegas2026], which consists of a Galerkin projection of the DG-FEM dynamics onto the subspace spanned by the coarse-grained states described by the average temperatures. The critical result of the coarse-graining procedure is that the LCM dynamics can be written in the following form,

4.2 Formulation of Reduced-Order Model

The Mori-Zwanzig (MZ) formalism is an operator-projection technique use to derive ROMs for high-dimensional dynamical systems, especially in statistical mechanics and fluid dynamics [Parish2017,Stinis2017,Parish
It provides an exact reformulation of the full-order dynamics in terms of a subset of resolved states. The proposed PIROM is subsequently developed based on such reformulation.

5 Bulk-only Mori–Zwanzig Memory Closure

The purpose of the closure is to account for the unresolved dynamics in the LCM, while leaving the contact-jump dynamics unchanged. Accordingly, the

Accordingly, write the Projected Generalized Langevin Equation (PGLE) for the lumped-temperature dynamics,

$$A(\mathbf{u})\dot{\mathbf{u}} = \sum_{i=1}^{N-1} \frac{1}{S_i(\mathbf{u})} \left(\mathbf{K}_i u + \mathbf{L}_i \Delta\boldsymbol{\theta}\right) + \mathbf{f}(t) + \int_{t_0}^{t} \boldsymbol{\kappa}(t, s, u, \mathbf{f}; \Theta) ds \tag{50}$$

Now, ensure the designed memory modifies only the bulk conductances, i.e.,

$$\boldsymbol{\kappa} \in \operatorname{span}\left\{\mathbf{K}_1 u(t), \mathbf{K}_2 u(t), \mathbf{K}_3 u(t), \mathbf{L}_1 \Delta\boldsymbol{\theta}_1(t), \mathbf{L}_2 \Delta\boldsymbol{\theta}_2(t), \mathbf{L}_3 \Delta\boldsymbol{\theta}_3(t)\right\} \tag{51}$$

which leads to,

$$\boldsymbol{\kappa}(t, s, u, \mathbf{f}; \Theta) \approx \sum_{j=1}^{m} \kappa_j(t, s, u, \mathbf{f}; \Theta) \boldsymbol{\mathcal{P}}_j \tag{52}$$

where,

$$\boldsymbol{\mathcal{P}}_j = \sum_{i=1}^{N-1} \frac{1}{S_i(u)} \mathbf{p}_{ij} \tag{53}$$

Now, define the memory variables as,

$$\beta_j(t) = \int_{t_0}^{t} \kappa_j(t, s, u, \mathbf{f}; \Theta) ds \tag{54}$$

then,

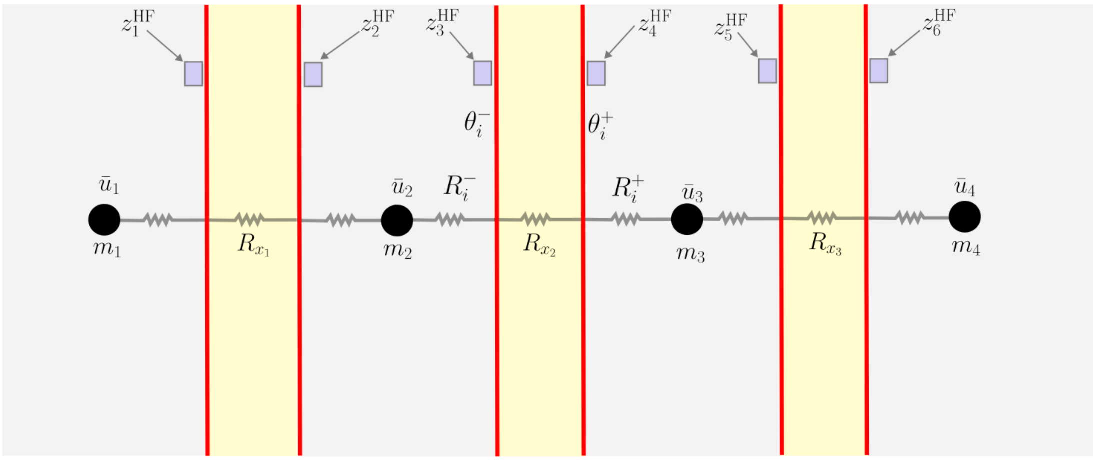
$$\int_{t_0}^{t} \boldsymbol{\kappa}(t, s, u, \mathbf{f}; \Theta) ds = \sum_{j=1}^{m} \beta_j(t) \boldsymbol{\mathcal{P}}_j \tag{55}$$

It follows that,

$$\sum_{j=1}^{m} \beta_j(t) \boldsymbol{\mathcal{P}}_j = \sum_{j=1}^{m} \beta_j(t) \sum_{i=1}^{N-1} \frac{1}{S_i(u)} \mathbf{p}_{ij} = \sum_{i=1}^{N-1} \frac{1}{S_i(u)} \left(\sum_{j=1}^{m} \mathbf{p}_{ij} \beta_j(t) \right) = \sum_{i=1}^{N-1} \frac{1}{S_i(u)} \mathbf{P}_i \boldsymbol{\beta} \tag{56}$$

Then, the memory correction to the LCM dynamics leads to the following modified lumped-temperature dynamics,

$$A(\mathbf{u})\dot{\mathbf{u}} = \sum_{i=1}^{N-1} \frac{1}{S_i(\mathbf{u})} \left(\mathbf{K}_i u + \mathbf{L}_i \Delta\boldsymbol{\theta} + \mathbf{P}_i \boldsymbol{\beta}\right) + \mathbf{f}(t) \tag{57}$$



$$q_i^{(c)} = \frac{\theta_i^+ - \theta_i^-}{R_{x_i}}$$

Now, let,

$$\kappa_i\left(t, s, \mathbf{u}, \mathbf{f}; \Theta\right) = e^{-\lambda_j(t-s)}\left(q_j^\top \phi(\mathbf{u}(s)) + r_j^\top \psi(\mathbf{f}(s))\right) \tag{58}$$

such that the memory variables are given by,

$$\boldsymbol{\beta}(t) = \int_{t_0}^{t} e^{-\Lambda(t-s)}\left(Q^\top \phi(\mathbf{u}(s)) + R^\top \psi(\mathbf{f}(s))\right) ds \tag{59}$$

The hidden dynamics are now given by,

$$\dot{\boldsymbol{\beta}} = -\Lambda\boldsymbol{\beta} + Q^\top\phi(\mathbf{u}) + R^\top\psi(\mathbf{f}) \tag{60}$$

and in state-space form, the hidden dynamics are,

$$\dot{\boldsymbol{\beta}} = -\Lambda\boldsymbol{\beta} + Q^\top\phi(\mathbf{u}) + R^\top\psi(\mathbf{f}) \tag{61}$$

The full lifted and closed system is given by,

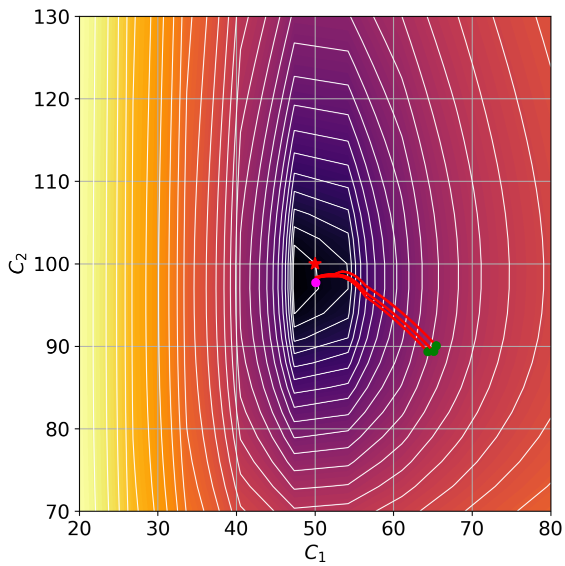
$$A(\mathbf{u})\dot{\mathbf{u}} = \sum_{i=1}^{N-1} \frac{1}{S_i(\mathbf{u})}\left(\mathbf{K}_i\mathbf{u} + \mathbf{L}_i\Delta\boldsymbol{\theta} + \mathbf{P}_i\boldsymbol{\beta}\right) + \mathbf{f}(t) \tag{62a}$$

$$M(\mathbf{u}; R_x)\dot{\Delta\boldsymbol{\theta}} = \mathbf{D}\dot{\mathbf{u}} - G_x\big(\Delta\boldsymbol{\theta} \odot \dot{S}(\mathbf{u})\big) \tag{62b}$$

$$\dot{\boldsymbol{\beta}} = -\Lambda\boldsymbol{\beta} + Q^\top\phi(\mathbf{u}) + R^\top\psi(\mathbf{f}) \tag{62c}$$

6 Application to Thermal Protection Systems

In this section, the proposed PIROM approach is applied to modeling a four-layered hypersonic TPS with unknown contact resistances, and is compared against a representative physics-based model, i.e., FEMs, and data-driven surrogate model, i.e., a neural ordinary differential equation (NODE) model. The performance of the models is assessed in terms of their ability to infer the TCRs, and to predict the transient temperature dynamics under varying boundary conditions and material properties. The configuration of the TPS, problem parametrization, optimal model design for inference, and numerical results are presented next.



6.1 Configuration of Thermal Protection System

6.2 Problem Parametrization

6.3 Optimal Model Design for Inference

This section presents the procedure for optimally selecting the physics-based model for inferring the TCR. Two critical design choices are discussed: (i) formulation of the physics-based model, and (ii) experimental limitations. In the first design choice, the resolution of the physics-based model is selected to balance the trade-off between model bias and computational cost. In the second design choice, experimental limitations, such as the number of thermocouples and their placement, are selected to ensure the identifiability of the TCR inference problem.

6.3.1 Resolution of the Physics-Based Model

The dimensionality of the physics-based model is critical for the performance of the PIROM, as it directly affects the computational cost and identifiability of the TCR inference problem. A coarse model may introduce undesirable model biases that can compromise the accuracy of the inferred TCRs, while a highly-refined model may introduce unnecessary computational costs that hinder efficiency during inference.

One-Dimensional Finite-Element Model

Lumped Capacitance Model

Lumped Capacitance Model with Jump Dynamics

6.3.2 Thermocouple Placement

In addition to the resolution of the physics-based model, the placement of the thermocouples is also critical for the performance of the PIROM, as it directly affects the identifiability of the TCR inference problem.

6.4 Numerical Results

7 Concluding Remarks

The main conclusion of this work are as follows,

1. A one-way physics-data PIROM design is found to successfully infer the unknown TCRs and hidden dynamics simulatneously. During inference, X and Y orders of magnitude in *training efficiency* are obtained relative to the FOM and 1D-FEM forward models, while X and Y orders of magnitude in *evaluation efficiency*

   X and Y orders of magnitude in *training efficiency* are obtained relati

   successfully infers the unknown TCRs and hidden dynamics simultaneously, offering X and Y orders of magnitude in *training efficiency*, as well as X and Y

   relative to the FOM and 1D-FEM based model, while X and Y orders of magnitude in *evlauation efficiency*

   and X and Y orders of magnitude in *evaluation efficiency*

   TCRs from noisy thermocouple

   framework successfully learns a ROM for the prediction of the temperature dynamics of the TPS, while simultaneously inferring the unknown contact conductances with errors of less than X% on noisy data with SNR of up to Y dB.

2. The PIROM outperforms the state-of-the-art data-driven, NN-based SciML, and traditional FEM modeling approaches for inferring contact conductances, as well as for predicting the temperature dynamics for OOD configurations.

3. Model bias is found to significantly affect the

8 Appendix

This appendix provides the technical details of the main text.

8.1 Lumped Capacitance Model

The TPS is divided into N components with M elements. For component i, define Ω_i as the spatial domain of the component, and $\partial\Omega_i$ as the boundary of the component. For each interface between two components, define $l(i)$ and $r(i)$ as the left and right components that form the interface, respectively.

For interface i, define $\Gamma_i = \partial\Omega_{l(i)} \cap \partial\Omega_{r(i)}$,

$$\theta_i^-(t) = \frac{1}{|\Gamma_i|}\int_{\Gamma_i} T_h^-(x,t)d\Gamma, \tag{63a}$$

$$\theta_i^+(t) = \frac{1}{|\Gamma_i|}\int_{\Gamma_i} T_h^+(x,t)d\Gamma \tag{63b}$$

$$\Delta\theta_i(t) = \theta_i^-(t) - \theta_i^+(t) \tag{63c}$$

Define the trace operators,

$$C_i^- \mathbf{u} = \frac{1}{|\Gamma_i|} \int_{\Gamma_i} \sum_{j=1}^{N_u} u_j(t) \phi_j(x) d\Gamma, \tag{64a}$$

$$C_i^+ \mathbf{u} = \frac{1}{|\Gamma_i|} \int_{\Gamma_i} \sum_{j=1}^{N_u} u_j(t) \phi_j(x) d\Gamma \tag{64b}$$

Stacking all the N_Γ interfaces, define the trace extraction operator as,

$$\mathbf{\Psi}^+ = \begin{bmatrix} C_1^- - C_1^+ \\ \vdots \\ C_{N_\Gamma}^- - C_{N_\Gamma}^+ \end{bmatrix} \in \mathbb{R}^{N_\Gamma \times MP} \tag{65}$$

where M is the number of elements and P is the number of basis functions per element. Therefore, it follows that,

$$\mathbf{s}(t) = \mathbf{\Psi}^+ \mathbf{u}(t) \tag{66}$$

Thus,

$$\mathbf{\Pi}^+ = \begin{bmatrix} \mathbf{\Phi}^+ \\ \mathbf{\Psi}^+ \end{bmatrix} \tag{67}$$

Introduce the lifting matrix,

$$\mathbf{\Psi} \in \mathbb{R}^{MP \times N_\Gamma} \tag{68}$$

such that,

$$\mathbf{\Phi}^+ \mathbf{\Psi} = \mathbf{0}, \quad \mathbf{\Psi}^+ \mathbf{\Psi} = \mathbf{I} \tag{69}$$

Note that $\Phi\bar{\mathbf{u}}$ contributes $E_\Gamma \bar{\mathbf{u}}$ to the interface jump, therefore the state decomposition can be written as,

$$\mathbf{u} = \mathbf{\Phi}\bar{\mathbf{u}} + \mathbf{\Psi}\left(\Delta\boldsymbol{\theta} - E_\Gamma \bar{\mathbf{u}}\right) + \delta\mathbf{u} = \tilde{\mathbf{\Phi}}\bar{\mathbf{u}} + \mathbf{\Psi}\Delta\boldsymbol{\theta} + \delta\mathbf{u} \tag{70}$$

where $\tilde{\mathbf{\Phi}} = \mathbf{\Phi} - \mathbf{\Psi} E_\Gamma$, and $\delta\mathbf{u}$ is the deviation from the average temperature in the components.